

Monadic second-order logic with cardinality predicates

OPG-37440

L1 - TRACTABLE

PARTIALLY SOLVED

Logic

ORIGINAL PROBLEM STATEMENT

The problem concerns the extension of Monadic Second Order Logic (over a binary relation representing the edge relation) with the following atomic formulas:

- " $\text{Card}(X) = \text{Card}(Y)$ "
- " $\text{Card}(X)$ belongs to A "

where A is a fixed recursive set of integers.

Let us fix k and a closed formula F in this language.

Conjecture Is it true that the validity of F for a graph G of tree-width at most k can be tested in polynomial time in the size of G ?